

ANRPC Action Tracker — Enterprise Production Documentation

Version: 1.0

System: ANRPC Action Tracker

Environment: Production / Enterprise Deployment

Owner: PSM / IT Department — ANRPC

1. Executive Summary

ANRPC Action Tracker is an enterprise-grade system designed to manage and monitor Process Safety Management (PSM) recommendations and actions across the refinery.

The system provides:

- Full lifecycle tracking (Source → Recommendation → Action)
- Role-based access control (RBAC)
- Centralized audit logging
- Secure authentication and authorization
- Operational dashboards and reporting
- Traceability and compliance readiness

The platform is designed to be:

- Secure
 - Auditable
 - Scalable
 - Maintainable
 - Production-ready
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2. System Objectives

The system supports the following business objectives:

- Track safety recommendations
- Assign responsibilities
- Monitor deadlines
- Control access permissions
- Record operational history
- Provide management visibility

- Support regulatory compliance
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3. System Architecture

High-Level Architecture

Frontend

React + TypeScript + Vite

Backend

Supabase

Components:

- PostgreSQL Database
- Authentication (Auth)
- Row-Level Security (RLS)
- Storage
- RPC Functions

Hosting

Netlify

Components:

- Frontend hosting
 - Serverless functions
 - API endpoints
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4. Core System Modules

4.1 Authentication Module

Features:

- User login
- User logout
- Password reset
- Temporary password

- Session management

Security:

- Supabase authentication
 - Token-based sessions
 - Secure password handling
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4.2 Users Management Module

Capabilities:

- Invite user
- Delete user
- Reset password
- Set password
- Assign role
- Activate / deactivate user

Administrative Controls:

- Admin-only operations
 - Role-based restrictions
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4.3 Roles & Permissions Module

The system uses Role-Based Access Control (RBAC).

Each role contains permissions.

Permissions control access to:

- Pages
 - Actions
 - API operations
 - Data access
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5. Roles & Permissions Matrix

Example Permissions:

Dashboard

- dashboard.view

Users

- settings.users.manage

Roles

- settings.roles.manage

Logs

- logs.view

Actions

- actions.create
- actions.update
- actions.close

Recommendations

- recommendations.create
- recommendations.update

Sources

- sources.create
 - sources.update
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6. Permission Enforcement Layers

Permissions are enforced in three layers.

Layer 1 — Frontend

React components check permissions before rendering.

Layer 2 — API

Netlify functions validate permissions.

Layer 3 — Database

Supabase RLS policies enforce security.

7. System Workflow

Core Workflow

Source

→ Recommendation

→ Action

→ Closure

Source Lifecycle

Create source

Assign category

Link recommendations

Track progress

Recommendation Lifecycle

Create recommendation

Assign priority

Assign responsible person

Track status

Action Lifecycle

Create action

Assign owner

Set due date

Update progress

Close action

8. Logs & Audit System

The system includes a full audit trail.

Logged Events

User authentication

Login

Logout

Password reset

System actions

Create source

Update source

Create recommendation

Update recommendation

Create action

Update action

Delete record

Role change

Permission change

Log Data Fields

Timestamp

User ID

User name

Action type

Entity

Entity ID

Description

Metadata

Logs Features

Pagination

Filtering

Search

Sorting

Role-based visibility

9. Dashboard Module

The dashboard provides operational visibility.

Dashboard Components

KPIs

Total actions

Open actions

Overdue actions

Closed actions

Charts

Priority distribution

Status distribution

Department performance

Owner workload

Due date timeline

10. Database Design

Core Tables

profiles

roles

role_permissions

sources

recommendations

actions

action_updates

audit_logs

Important Relationships

One Source

→ Multiple Recommendations

One Recommendation

→ Multiple Actions

One User

→ One Role

One Role

→ Multiple Permissions

11. API & RPC Functions

Core RPC Functions

set_role_permissions

has_role_permission

write_audit_log

reserve_source_number

reserve_recommendation_number

reserve_action_number

12. Netlify Functions

Serverless API Endpoints

admin-invite-user

admin-set-password

admin-reset-password

admin-delete-user

write-audit-log

13. Security Model

Authentication

Token-based sessions

Secure password storage

Authorization

Role-based permissions

Database RLS policies

Data Protection

HTTPS encryption

Environment variable protection

Restricted API access

14. Environment Variables

Frontend

VITE_SUPABASE_URL

VITE_SUPABASE_ANON_KEY

VITE_APP_URL

Backend

SUPABASE_SERVICE_ROLE_KEY

APP_URL

15. Deployment Guide

Step 1 — Database Setup

Run SQL migrations.

Step 2 — Deploy Functions

Deploy Netlify functions.

Step 3 — Build Application

Run:

```
npm run build
```

Step 4 — Deploy Production

Deploy to Netlify.

16. Backup & Recovery Strategy

Database Backup

Daily automatic backup

Manual backup before updates

Recovery Plan

Restore database snapshot

Restart services

Verify system integrity

17. Monitoring & Maintenance

System Monitoring

Error tracking

Performance monitoring

Log monitoring

Maintenance Tasks

Review logs

Clean inactive users

Verify permissions

Update dependencies

18. User Guide — How to Use the System

Step 1 — Login

Enter username

Enter password

Click Login

Step 2 — Create Source

Open Sources page

Click Create

Enter details

Save

Step 3 — Create Recommendation

Select source

Enter recommendation

Save

Step 4 — Create Action

Select recommendation

Enter action details

Assign responsible

Set due date

Save

Step 5 — Monitor Progress

Open dashboard

Review status

Track overdue items

19. Administrator Guide

Administrators can:

Create users

Assign roles

Modify permissions

Reset passwords

Delete users

View logs

20. Troubleshooting Guide

Issue — Cannot Login

Check password

Check user status

Reset password

Issue — Permission Denied

Check role permissions

Reload session

Verify role assignment

Issue — Logs Not Visible

Check logs.view permission

Verify database connection

Check RLS policies

21. Performance Best Practices

Use pagination

Avoid large queries

Index critical fields

Monitor slow queries

22. Change Management Process

All system changes must follow:

Development

Testing

Approval

Deployment

Verification

23. Compliance & Audit Readiness

The system supports:

PSM compliance

Audit traceability

User accountability

Change tracking

24. Future Enhancements

Planned features:

Email notifications

Mobile optimization

Advanced reporting

Data export

Integration with DCS

25. System Status — Current State

The system is:

Stable

Secure

Role-driven

Fully auditable

Production-ready

26. Architecture Diagram (Logical)

User Browser

→ Frontend (React Application)

→ Netlify Hosting

→ Netlify Functions (API Layer)

→ Supabase Services

- Authentication
- PostgreSQL Database
- Storage
- RPC Functions

Security Layers:

User → Authentication → Authorization → Database RLS → Audit Logs

27. Database ERD (Entity Relationship Overview)

Core Entities:

profiles

roles

role_permissions

sources

recommendations

actions

action_updates

audit_logs

Relationships:

profiles.role_id

→ roles.id

roles.id

→ role_permissions.role_id

sources.id

→ recommendations.source_id

recommendations.id

→ actions.recommendation_id

users.id

→ audit_logs.user_id

28. Roles & Permissions Matrix

Role	Dashboard	Users	Roles	Logs	Sources	Recommendations	Actions
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Admin	Full	Full	Full	Full	Full	Full	Full
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PSM Manager	View	Manage	View	View	Manage	Manage	Manage
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Department User	View	No	No	No	View	View	Update
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Viewer	View	No	No	No	View	View	View
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Permission Levels:

View

Create

Update

Delete

Manage

29. API Endpoints Documentation

Authentication APIs

POST /login

POST /logout

POST /reset-password

User Management APIs

POST /.netlify/functions/admin-invite-user

POST /.netlify/functions/admin-set-password

POST /.netlify/functions/admin-reset-password

POST /.netlify/functions/admin-delete-user

Logs APIs

POST /.netlify/functions/write-audit-log

GET /logs

30. Operational SOP (Standard Operating Procedure)

Daily Operations

Login to system

Review dashboard

Check overdue actions

Update action status

Verify assignments

Weekly Operations

Review logs

Verify permissions

Validate data accuracy

Monthly Operations

Backup verification

Performance review

Security audit review

31. User Manual — Step-by-Step Usage

Step 1 — Login

Open system URL

Enter email

Enter password

Click Login

Step 2 — Create Source

Go to Sources page

Click Create Source

Enter source details

Save

Step 3 — Create Recommendation

Select source

Enter recommendation details

Save

Step 4 — Create Action

Select recommendation

Enter action details

Assign responsible

Set due date

Save

Step 5 — Update Action

Open action record

Enter update details

Save

Step 6 — Close Action

Verify completion

Update status to Closed

Save

32. Go-Live Checklist

System Configuration

Database migration completed

Roles configured

Permissions configured

Users created

Email settings configured

Security

HTTPS enabled

Environment variables secured

RLS policies enabled

Operational

Backup configured

Monitoring enabled

Logs verified

Performance tested

33. System Maintenance Schedule

Daily

Check system availability

Monitor errors

Review logs

Verify login functionality

Weekly

Review user access

Check overdue actions

Verify backups

Update dependencies if required

Monthly

Security review

Performance review

Database optimization

Audit log review

Quarterly

Full system audit

Disaster recovery test

Permission review

Infrastructure health check

End of Document